Religious Coping in Times of Crisis: Evidence from COVID-19 Lockdowns and Google Trends Data

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February 2026

Abstract

This paper studies how religious interest evolved around COVID-19 lockdown announcements in the US using high-frequency Google Trends data. Leveraging the staggered timing of state-level lockdowns during the first 100 days of 2020, and pairing each calendar day with its counterpart in 2019, I employ a difference-in-differences design to examine changes in religious search behavior. I find a substantial increase in religion-related searches following lockdown announcements. Topics such as Prayer increase by close to one standard deviation, a magnitude comparable to that observed for mental health–related topics. The rise in searches for God suggests not only greater search intensity among already religious users, but also increased engagement by people who do not regularly participate in religious practices. These coping responses begin to emerge roughly one week prior to lockdown announcements, likely reflecting heightened information flows and growing public awareness that restrictive measures were forthcoming.

1 Introduction

The COVID-19 pandemic triggered an abrupt and unprecedented disruption to everyday economic and social life around the world. Although no nationwide lockdown was implemented in the US, most states enforced lockdown policies that restricted mobility, suspended in-person work and schooling, and sharply reduced social interaction. Beyond their economic consequences, these policies generated a sudden rise in uncertainty, anxiety, and psychological distress. In such contexts, religion can serve as a source of meaning, reassurance, and emotional support.

A growing literature documents the role of religion during the COVID-19 pandemic, particularly in relation to mental health and coping. David et al. (2023) conduct a meta-analysis covering the first year of the pandemic and find that religion was frequently reported as a coping mechanism, often with favorable associations with mental health outcomes. Using survey data from the US, Bahal et al. (2023) show that COVID-19 infection was associated with worse mental health, but that this association was significantly weaker among religious respondents. Evidence from the United Kingdom points in a similar direction. Based on an online survey of regular worshipers, Baggaley et al. (2024) document substantial deterioration in mental health during lockdowns, while nearly all respondents reported that prayer helped them cope with the crisis. Beyond survey evidence and national case studies, global digital trace data tell a similar story. Using Google search data from 107 countries, Bentzen (2021) documents a sharp and unprecedented rise in searches for prayer during the early months of the pandemic, suggesting a widespread increase in religious engagement as individuals sought to cope with uncertainty and distress.

While this body of evidence highlights a close link between religion and psychological well-being during the pandemic, it is largely correlational. In contrast, Brodeur et al. (2021) provide causal evidence on the mental health consequences of COVID-19 lockdowns in both Europe and the US. Using Google Trends data, they show that lockdown policies led to sharp increases in Google searches related to boredom, loneliness, worry, and sadness, showing a clear deterioration in mental well-being following lockdowns. Taken together, these findings suggest that lockdowns constituted a sudden psychological and social shock, and that religion may have played a role in how individuals responded. Just as households rely on economic coping mechanisms in response to adverse economic events, individuals may also turn to psychological or social coping mechanisms when faced with stress, uncertainty and isolation. This raises a natural question that has received little attention so far: did COVID-19 lockdowns causally affect religious engagement?

Using daily data for all 50 US states, I examine how searches related to religion evolved around state-level lockdown announcements during the first 100 days of 2020. To control for seasonality and the timing of religious holidays, each day in 2020 is paired with the corresponding calendar day in 2019. I analyze six broad Google Trends topics (Faith, God, Meditation, Prayer, Religion, and Spirituality) and exploit the staggered timing of lockdown announcements within a difference-in-differences (DiD) framework. I estimate a a two-way fixed effects (TWFE) model similar to Brodeur et al. (2021), and in addition use the DiD imputation estimator proposed by Borusyak et al. (2024). These designs leverage both cross-state variation in policy timing and within-state comparisons across years.

I find a sizable increase of close to one standard deviation in searches for Prayer, providing further evidence of its role as a coping mechanism and aligning with findings in Baggaley et al. (2024) and Bentzen (2021). Related concepts such as Faith, God, and Meditation also yield positive and statistically significant estimates. Although Google Trends data alone do not allow for a distinction between intensive and extensive margin responses, the findings are unlikely to be explained solely by increased activity among already religious individuals. In particular, prior work shows that Google searches for God are strongly correlated with the share of the US population that believes in God (Stephens-Davidowitz (2014)). The positive effects observed for God therefore suggest a broader increase in religious belief rather than a mere intensification among the already religious. The absence of corresponding increases in searches for Religion or Spirituality further suggests that the response was oriented toward personal and immediate forms of religious meaning rather than abstract or institutional religion.

This pattern of results sheds light on the behavioral mechanisms underlying religious responses to crisis. By documenting how religious interest responded in real time to lockdown announcements, this paper contributes to the literature on beliefs, coping behavior, and individual responses to large-scale social shocks. More broadly, it shows how high-frequency digital trace data, such as daily Google Trends series, can be used to capture changes in public interest and behavior following major public interventions.

The remainder of the paper is organized as follows. Section 2 describes the Google Trends data, the religious topics considered, and the construction of the sample. Section 3 presents the empirical strategy used to identify the causal effect of lockdowns on religiosity. Section 4 discusses the main findings. Section 5 concludes.

2 Data

2.1 Google Trends Data

Religiosity is measured using data from Google Trends, which provides aggregated information on the relative search intensity of selected keywords or topics. Rather than reporting raw search counts for a given region and time period, Google Trends reports a normalized 0–100 Relative Search Index[1] (RSI), constructed by dividing the number of searches at each point in time by the maximum search volume observed over the sample period and rescaling. Topic-based queries, unlike keyword queries, are cross-language and incorporate misspellings, acronyms, and conceptually related terms. As a result, topic queries provide a more comprehensive and stable measure of religiosity than exact-string keyword searches.

As noted by Hölzl et al. (2025), Google Trends data present both advantages and challenges for empirical research. Because search behavior is generated as part of users' routine online activity and does not require active participation in a survey, it is not subject to cognitive biases such as social desirability bias, recall error, or misinterpretation of survey questions. Moreover, search-based measures do not depend on respondents' willingness to provide thoughtful or honest answers. Relative to survey-based measures, such as those derived from Gallup or similar instruments, Google Trends data offer substantially higher temporal frequency and broad geographic coverage, allowing for the analysis of short-run dynamics and responses to concrete events that are difficult to capture in infrequent surveys[2].

At the same time, an important challenge in the use of Google Trends data concerns internal validity, namely whether the selected queries capture the underlying construct of interest. While search activity measures revealed interest in a topic, it may not be an adequate measure for endorsement, behavior, or knowledge. For example, a user searching for information on jihadist terrorism is not necessarily expressing support for it. In that sense, one can confidently use Google Trends data to measure public attention, whereas any interpretation beyond interest requires additional justification. Results should therefore be interpreted primarily as changes in public interest and only when carefully argued as changes in underlying attitudes or behavior.

External validity also warrants consideration, as Google Trends data reflect the behavior of Google users rather than the entire population. This concern is mitigated for the US since Google usage is widespread. Finally, Google Trends reports unbiased point estimates derived from a random sample of search queries to which a 0-100 normalization procedure is applied. Although Google does not disclose details of the sampling algorithm, sampling error won't threat identification as long as it is orthogonal to the covariates.

[1] This term is used by Hölzl et al. (2025), however there is no official term to refer to this scale

[2] Yeung (2019) uses Google Trends data to construct an index for Christian Religiosity, but isn't well suited for daily-frequency variation.

2.2 Sample and topics selection

I consider six broad Google Trends topics to proxy for religiosity: (1) Faith, (2) God, (3) Meditation, (4) Prayer, (5) Religion, and (6) Spirituality. For each topic, I estimate specifications using the unrestricted query, although as well as versions filtered by the category 'Religion & Belief', which restricts the underlying searches to those Google classifies as religious in nature and thereby improves measurement relevance. The category 'Religion & Belief' is further divided into fourteen subcategories that capture more specific religious domains. I analyze each of these subcategories separately, including those related to Christianity, Islam, and Judaism, with each subcategory aggregating searches that Google assigns to that particular religious domain.

To study the effect of lockdowns on religious interest, I focus on the first 100 days of 2020 for all 50 US states. Each day in 2020 is paired with the corresponding calendar day in 2019, which controls for seasonality and for the timing of religious holidays relative to lockdown implementation. However, since each daily Google Trends series is normalized relative to the maximum search intensity within its own query window, an index value of 100 in 2019 is not directly comparable to a value of 100 in 2020. To compare daily Google Trends (GT) series that differ in their Relative Search Index (RSI) scales, I follow the normalization approach proposed by Caporin and Poli (2017) which I describe next.

Let $(d_t)_{t\in T}$ and $(d'_t)_{t\in T'}$ denote the daily GT series for 2020 and 2019, respectively, and let $(w_i)_{i\in I}$ represent a weekly series that spans all days included in both daily series. The goal is to obtain two directly comparable daily series each with one hundred days. The process consists of three main steps[3]:

1. **Weekly normalization.** Each daily observation is expressed relative to the total GT volume of its corresponding week:
$$d_{t,rel} = \frac{d_t}{\sum_{j=t_1}^{t_7} d_j} \ , \ \ d'_{t,rel} = \frac{d'_t}{\sum_{j=t_1}^{t_7} d'_j}$$
Since each week contains seven days, division by the weekly average would yield an equivalent normalization.

2. **Weekly rescaling.** Each relative daily value is then multiplied by its corresponding weekly value, resulting in two series that are now comparable across years:
$$c_t = d_{t,rel} \times w_{i(t)} \ , \ \ c'_t = d'_{t,rel} \times w_{i(t)}$$

3. **RSI adjustment.** Finally, both comparable series are rescaled to align with the original RSI range:
$$\tilde{d}_t = \frac{c_t}{\max\{(c_t)_{t\in T}, (c'_t)_{t\in T'}\}} \ , \ \ \tilde{d}'_t = \frac{c'_t}{\max\{(c_t)_{t\in T}, (c'_t)_{t\in T'}\}}$$

The resulting series $\tilde{d}_t$ and $\tilde{d}'_t$ are directly comparable and take values between 0 and 100. Combining them yields a single series of length $|T| + |T'|$, from which I retain 100 days per year as intended. Since the first (last) seven days of the target 100-day window may not align exactly with the start (end) of a given week, I extend the original daily series to include additional days for proper weekly normalization. In this case, the required sets are $T = \{-3, -2, -1, 1, \ldots, 101\}$ and $T' = \{-2, -1, 1, \ldots, 103\}$, where day 1 corresponds to the first day of the year and day -1 to the last day of the preceding year. Specific dates and additional days considered for proper weekly normalization are specified in Figure 1.

3 Empirical Strategy

3.1 Twoway Fixed Effects and Imputation

The empirical strategy leverages the panel structure of the Google Trends data and the staggered implementation of COVID-19 lockdowns within a difference-in-differences framework. The daily frequency of the

[3] This procedure can be readily generalized to compare more than two daily series or to accommodate data measured weekly. In the latter case, one simply replaces the weekly reference series with an adequate monthly series to maintain consistency.

series enables us to incorporate 2019 observations to construct the counterfactual trajectory, in addition to not-yet-treated units, as is standard DiD designs. Similar to Brodeur et al. (2021), I estimate the following model two-way fixed effects (TWFE) model,

$$y_{it} = \gamma \mathcal{D}_{it} \times \mathbb{1}_{2020} + \delta \mathcal{D}_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{1}$$

where y_{it} are Google searches for a specific topic of interest in state i at time t, $\mathbb{1}_{2020}$ an indicator variable for year 2020, $\mathcal{D}$ is a dummy equal to one since the day lockdown is announced in 2020 and its paired day from 2019, and μ_i and λ_t are state and time fixed effects respectively. Time fixed effects consist of day-of-year (1-100), day-of-week (dow), and year fixed effects[4]. The parameter of interest is γ, standard errors are clustered at the day-of-year level, and regressions are weighed by state population.

This specification differs from a standard staggered-adoption TWFE model in how pre-treatment information from 2019 is incorporated into the model. Given the definition of $\mathcal{D}$, the interaction $D_{it} = \mathcal{D}_{it} \times \mathbb{1}_{2020}$ is equivalent to the usual post-treatment indicator in a TWFE design, while controlling for $\mathcal{D}$ controls for being in the same time of year since lockdown was announced but for the paired 2019 days. Also, given that this design relies on within-state comparisons across paired days in 2019 and 2020, I use of day-of-year fixed effects instead of calendar-date fixed effects, since the latter would eliminate the cross-year variation required for these comparisons. In addition, I include day-of-week fixed effects to account for systematic weekly patterns in religious search activity, since paired days across years do not generally align on the same day of the week and religious interest tends to peak on weekends.

I also adopt the imputation estimator of Borusyak et al. (2024) to overcome limitations of standard TWFE and event-study regressions in staggered designs with heterogeneous effects, which may combine effects with negative or non-transparent weights and contaminate dynamic patterns (De Chaisemartin and d'Haultfoeuille (2020); Callaway and Sant'Anna (2021); Sun and Abraham (2021)). Using similar notation, within the set of observations $it \in \Omega$, I define the set of treated observations as $\Omega_1 = \{it \in \Omega : D_{it} = 1\}$ and $\Omega_0 = \{it \in \Omega : D_{it} = 0\}$ as the set of untreated observations. I estimate the effect of Covid-19 lockdowns in three steps:

1. For untreated observations $it \in \Omega_0$ and weighting by population (w_i), regress $y_{it} = \delta \mathcal{D}_{it} + \mu_i + \lambda_t + \varepsilon_{it}$
2. For treated observations $it \in \Omega_1$, compute $\hat{\tau}_{it} = y_{it} - \hat{y}_{it}(0)$, where the last term comes from extrapolating the previous model to the treated subsample
3. Aggregate for treated observations using the weighted average $\hat{\tau}_w = \sum_{it \in \Omega_1} w_i \hat{\tau}_{it}$

This imputation approach only models untreated potential outcomes, while allowing treatment effects to vary freely across units and over time. It is therefore well suited to the time fixed-effects structure of the TWFE model, and the 2019 observations naturally contribute to the construction of the counterfactual. In addition, the approach permits highly flexible aggregation of treatment effects. By averaging the estimated treatment effects $\hat{\tau}_{it}$ over seven-day periods following the announcement, I leverage the statistical power of the daily data while preserving day-of-week fixed effects[5]. For both the TWFE and Imputation estimator never-treated states are not considered, although including them yield similar estimates.

3.2 Identification concerns

Identification relies on a parallel-trends assumption: absent lockdowns, outcomes in treated states would have evolved similarly to a year ago and to not-yet-treated states in 2020. To assess the plausibility of this assumption, I follow the pre-trends testing procedure proposed by Borusyak et al. (2024), which includes leads of treatment status in the first-stage regression of the imputation estimator. Specifically, I estimate the following model using pre-treatment data $it \in \Omega_0$.

[4]This follows a more flexible structure than the one specified in Brodeur et al. (2021), which considers week-of-year instead of day-of-year.

[5]This would not be possible if observations were collapsed to the weekly level or into non-overlapping seven-day periods relative to announcement.

$$y_{it} = \sum_{k=-11}^{-1} \tau_k \mathcal{E}_{it}^k \times \mathbb{1}_{2020} + \sum_{k=-11}^{-1} \delta_k \mathcal{E}_{it}^k + \delta \mathcal{D}_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{2}$$

Similarly to $\mathcal{D}$, $\mathcal{E}^k$ are dummy variables equal to one k weeks relative to the lockdown announcement, defined for 2020 and the 2019 paired days. The omitted category is twelve or more weeks prior to lockdown announcement. Under the parallel-trends assumption $E[y_{it}(0)] = \delta\mathcal{D}_{it} + \mu_i + \lambda_t$ for all $i, t \in \Omega$, hence statistically significant estimates of τ_k would indicate differential pre-treatment trends and evidence against this identifying assumption.

Another identification concern is that changes in search intensity may be driven by the evolution of the COVID-19 pandemic itself rather than by lockdown policies. To address this concern, Figure 3 compares TWFE estimates from specifications that include controls for local COVID-19 severity, such as one-day lagged confirmed deaths. The estimated effects are very similar when additionally controlling for lagged confirmed cases or for moving averages of lagged deaths and cases. These results suggest that the estimated effects are not primarily driven by contemporaneous pandemic dynamics, but rather by the implementation of lockdown policies.

COVID-19 controls are also not included in the first-stage regression of the imputation estimator. The lack of variation in pre-treatment and, most importantly, the absence of higher values for not-yet-treated observations, makes the 'out-of-sample' predictions unreliable when extrapolating for treated observations.Reassuringly, the similarities when controlling for them in TWFE estimates indicate that omitting these controls from the imputation procedure is unlikely to materially affect the results.

4 Results

4.1 Topic evolution and DiD Estimates

Figure 2 plots the evolution of topic popularity before and after the lockdown announcement by year, after adjusting series from different time frames to be comparable. A noticeable increase in search intensity emerges for most topics roughly 10 to 7 days prior to the announcement of state lockdowns. For Faith, God, Meditation, and Prayer, trends are closely aligned with their 2019 counterparts up to this point, after which search interest rises sharply relative to 2019. In the case of Prayer, search intensity exhibits a second, pronounced increase approximately two weeks after lockdown announcement. In contrast, Religion and Spirituality display different patterns. Searches for Religion follow broadly similar trajectories before and after the announcement, except for a temporary decline in the weeks preceding the lockdown. Searches for Spirituality begin to diverge from their 2019 counterparts more than two months before the announcement, which may be indicative of differential pre-period dynamics for this topic.

Table 1 reports estimates of the effect of lockdown announcements obtained using both the TWFE specification and the imputation estimator. The two approaches yield very similar results. Large and positive effects are estimated for Faith, God, Meditation, and Prayer, while no statistically significant effects are found for Religion, and point estimates for Spirituality are small in magnitude but statistically significant. I find sizable significant effects for prayer in particular. The lockdown announcement increases Prayer's RSI by 18.12 and 20 points in the TWFE and BJS specifications, respectively. These results are consistent with survey-based evidence highlighting prayer as a central coping response during the pandemic (Baggaley et al., 2024), and with prior work documenting sharp increases in prayer-related searches during COVID (Bentzen, 2021).

The magnitude of these effects is economically meaningful. The estimated increases correspond to nearly 40 percent of the pre-lockdown mean and imply changes of 0.86 and 0.95 standard deviations. These effects are comparable in size to the increase in boredom searches following lockdowns in the US documented by Brodeur et al. (2021), and substantially larger than the corresponding effects on loneliness, worry, and sadness. Taken together, these comparisons suggest that lockdowns not only generated sizable negative impacts on mental well-being, but also induced large behavioral responses in the form of increased prayer. This pattern is consistent with prayer functioning as an active coping response to the psychological and social disruption caused by the lockdowns and the broader crisis.

Figure 5 presents TWFE estimates for each subcategory within the 'Religion & Belief'. The strongest positive associations appear for religious traditions that are most prevalent in the US. Searches related to Judaism and Christianity are higher by 15.43 and 13.80 RSI points. In contrast, lockdowns are negatively associated with searches related to Theology. This pattern suggests a shift in religious interest away from abstract or doctrinal topics toward more concrete practices, such as prayer, and toward religious traditions with broader and more established communities in the US. It is worth noting that these estimates should not be interpreted causally. Pre-trends tests reveal substantial differential trends prior to lockdown announcements, likely reflecting the broad nature of these subcategories, which aggregate a wide range of topics and are therefore more susceptible to being influenced by other contemporaneous events within the 12-week window.

4.2 Pre-trends test

Figure 4 plots the estimated coefficients τ_k from equation 2. Consistent with the graphical analysis in figure 2, this figure suggests that the pre-treatment dynamics for God and Prayer are generally well behaved, with the exception of the week immediately preceding the lockdown announcement[6]. For that corresponding week, the estimated coefficients indicate increases of approximately 7 and 16 RSI points for God and Prayer, respectively. This short deviation is plausibly attributable to increased news circulation and growing public awareness that state-level lockdowns were likely to be implemented in the near future.

When treatment is redefined as beginning seven days prior to announcement, this pre-announcement increase is substantially attenuated, and the coefficient for the week immediately preceding treatment becomes only marginally different from zero for God, Prayer, and Meditation. In contrast, Faith, Religion, and Spirituality continue to exhibit differential pre-treatment dynamics. Table 2 reports TWFE and BJS estimates using the redefined treatment timing. The estimated effects for God and Prayer remain very similar, declining by at most about one RSI point. In contrast, the estimated effect for Meditation is more sensitive to this redefinition, falling from roughly 10 RSI points to about 7 RSI points.

5 Conclusion

This paper studies how religious interest responded to COVID-19 lockdown announcements in the US using daily Google Trends data. Exploiting the staggered timing of state-level lockdowns and pairing 2020 observations from with the corresponding calendar days in 2019, I document a pronounced rise in searches related to religion, most notably prayer, following lockdown announcements. Estimates point to large and economically meaningful increases in prayer-related searches, on the order of nearly one standard deviation.

These findings contribute to a growing literature on how individuals respond to large-scale social and psychological shocks. While prior work has shown that lockdowns generated sharp deterioration in mental well-being (Brodeur et al., 2021), these results suggest that individuals also engaged in active coping responses. The increase in Prayer searches aligns with survey-based evidence documenting the use of prayer as a coping mechanism during the pandemic (Baggaley et al., 2024) and complements global evidence of increased religious engagement during COVID-19 (Bentzen, 2021). On the other hand, increase in topic of God signals an increase of religiosity in the extensive margin given its high correlation with the share of US population that believes in God Stephens-Davidowitz (2014). Finally, the evidence suggests that these coping responses start to materialize about one week before lockdown announcements, consistent with increased information flows and growing public awareness of the evolving severity of the pandemic.

[6] This pattern remains unchanged when COVID-19–related controls are included in the pre-trends regression.

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6 Tables

Table 1: DiD estimates on Topics

	Faith (1)	God (2)	Meditation (3)	Prayer (4)	Religion (5)	Spirituality (6)
I. TWFE Estimates						
ATT (γ)	13.564***	9.809***	9.189***	18.116***	0.972	-2.525**
	(1.171)	(0.567)	(1.601)	(1.338)	(1.600)	(1.249)
Mean: pre-lockdown	44.12	63.54	45.83	49.55	54.37	47.99
p-value	0.000	0.000	0.000	0.000	0.545	0.046
Observations	6,363	8,388	7,180	8,131	8,117	6,673
II. BJS Estimates						
ATT (τ)	15.592***	10.754***	9.969***	20.001***	1.155	-2.808***
	(1.036)	(0.562)	(0.641)	(0.780)	(1.129)	(0.837)
p-value	0.000	0.000	0.000	0.000	0.306	0.001
Observations in Ω_0	5,829	7,720	6,572	7,475	7,457	6,123
Observations in Ω_1	534	668	608	656	660	550

Notes: This table reports TWFE (Panel I) and BJS (Panel II) estimates for Google Trends topics. State, year, day-of-year and dow fixed effects are included as specified in equation 1. Clustered standard errors at the day level in parenthesis. *signifcantant 10%, **signicantat 5%,***signicantant 1%

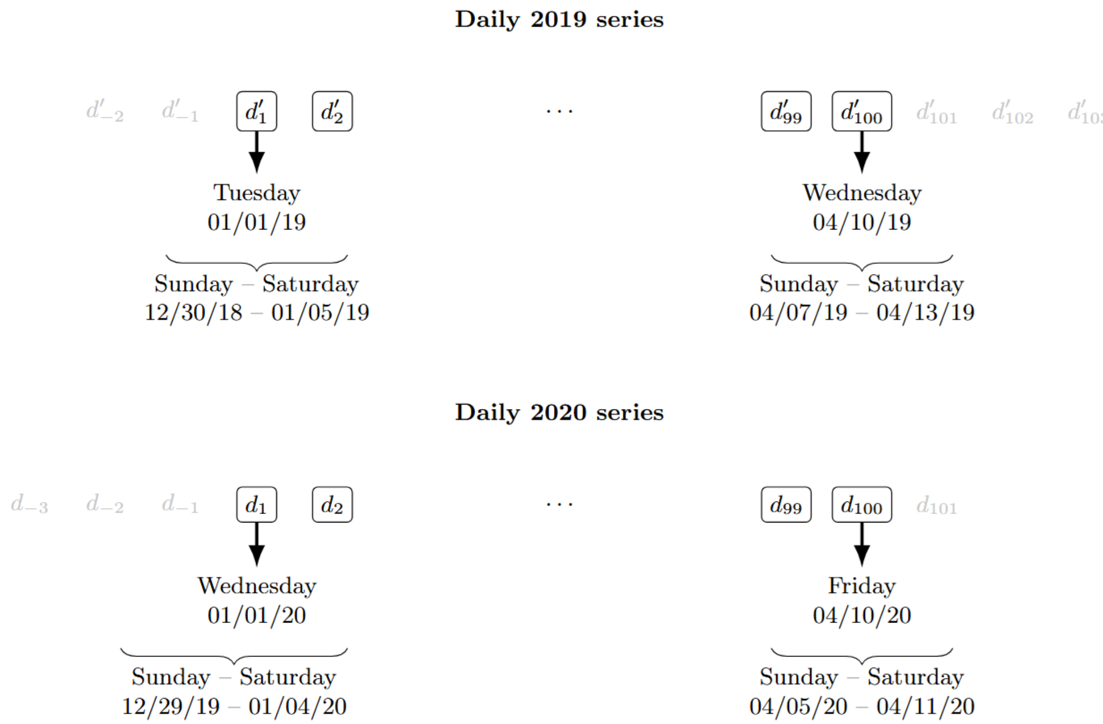
Table 2: DiD estimates on Topics (1-week anticipation)

	Faith (1)	God (2)	Meditation (3)	Prayer (4)	Religion (5)	Spirituality (6)
I. TWFE Estimates						
ATT (γ)	12.629***	9.142***	6.738***	18.258***	-4.782***	-5.653***
	(1.134)	(0.753)	(1.417)	(1.142)	(1.700)	(1.244)
Mean: pre-lockdown	43.90	63.33	45.90	48.98	54.82	48.13
p-value	0.000	0.000	0.000	0.000	0.006	0.000
Observations	6,363	8,388	7,180	8,131	8,117	6,673
II. BJS Estimates						
ATT (τ)	14.462***	9.721***	6.840***	19.241***	-4.206***	-5.799***
	(0.909)	(0.455)	(0.578)	(0.602)	(0.914)	(0.819)
p-value	0.000	0.000	0.000	0.000	0.000	0.000
Observations in Ω_0	5,602	7,426	6,305	7,188	7,179	5,894
Observations in Ω_1	761	962	875	943	938	779

Notes: This table reports TWFE (Panel I) and BJS (Panel II) estimates for Google Trends topics. Treatment is redefined to start 7 days before announcement. State, year, day-of-year and dow fixed effects are included as specified in equation 1. Clustered standard errors at the day level in parenthesis. *signifcantant 10%, **signicantat 5%,***signicantant 1%

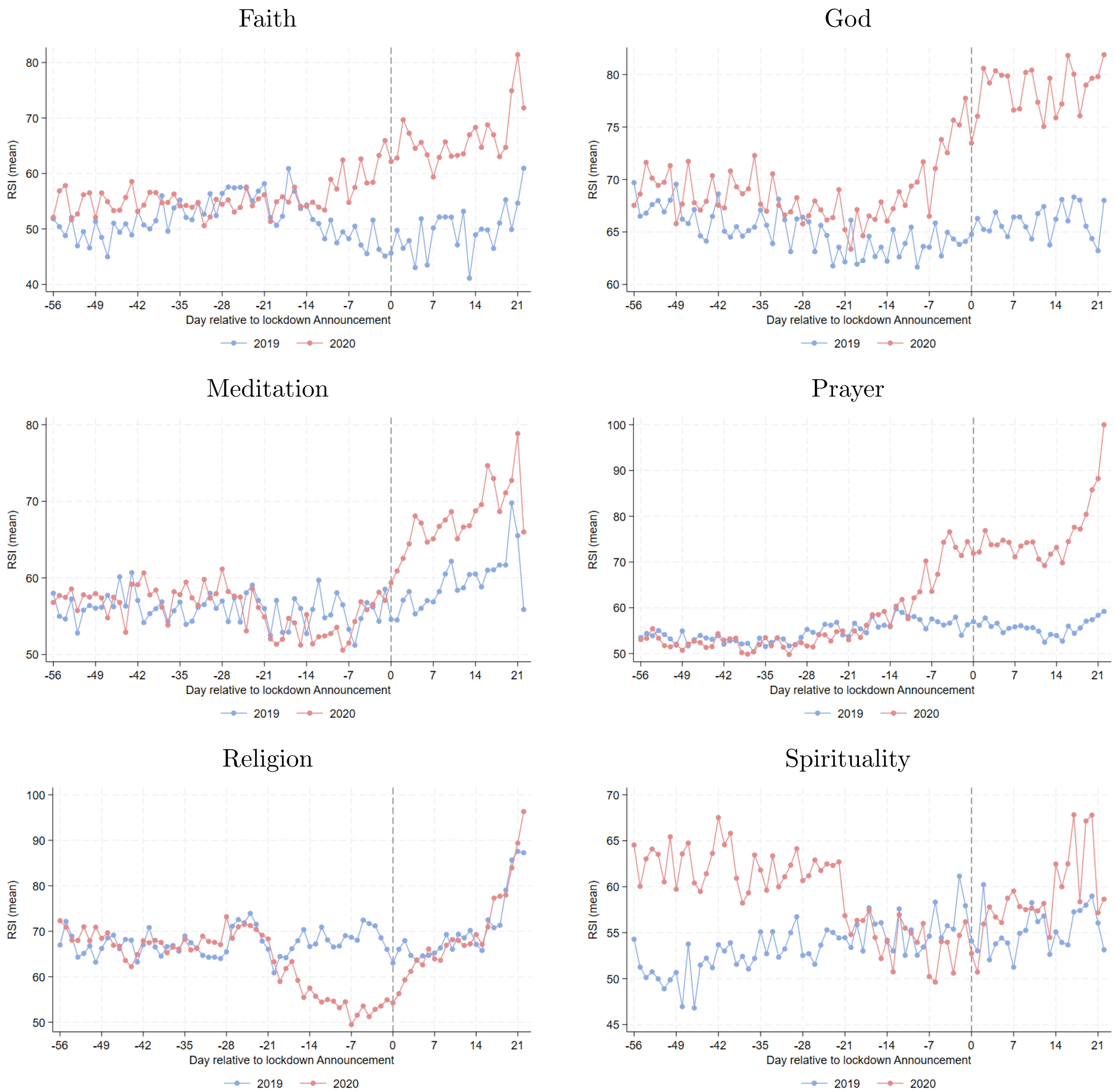
7 Figures

Figure 1: Sample selection and timeline



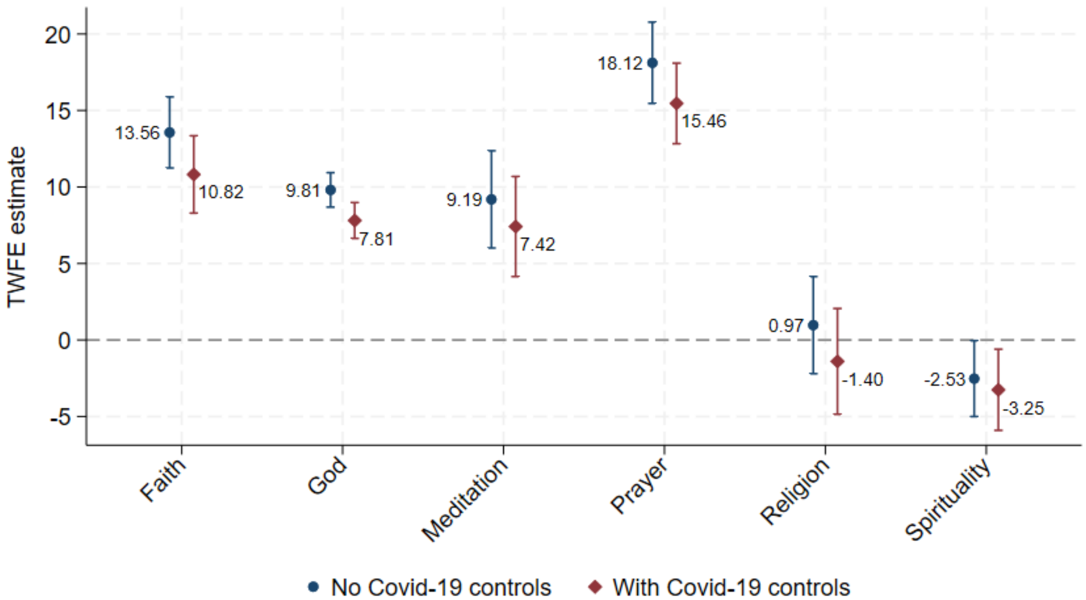
Notes: This figure shows the exact dates considered for analysis and rescaling process, for both 2019 and 2020 timelines. Days that are not included for regression analysis but are necessary for a proper weekly normalization process are displayed in gray.

Figure 2: Topic Series: 2019 vs. 2020



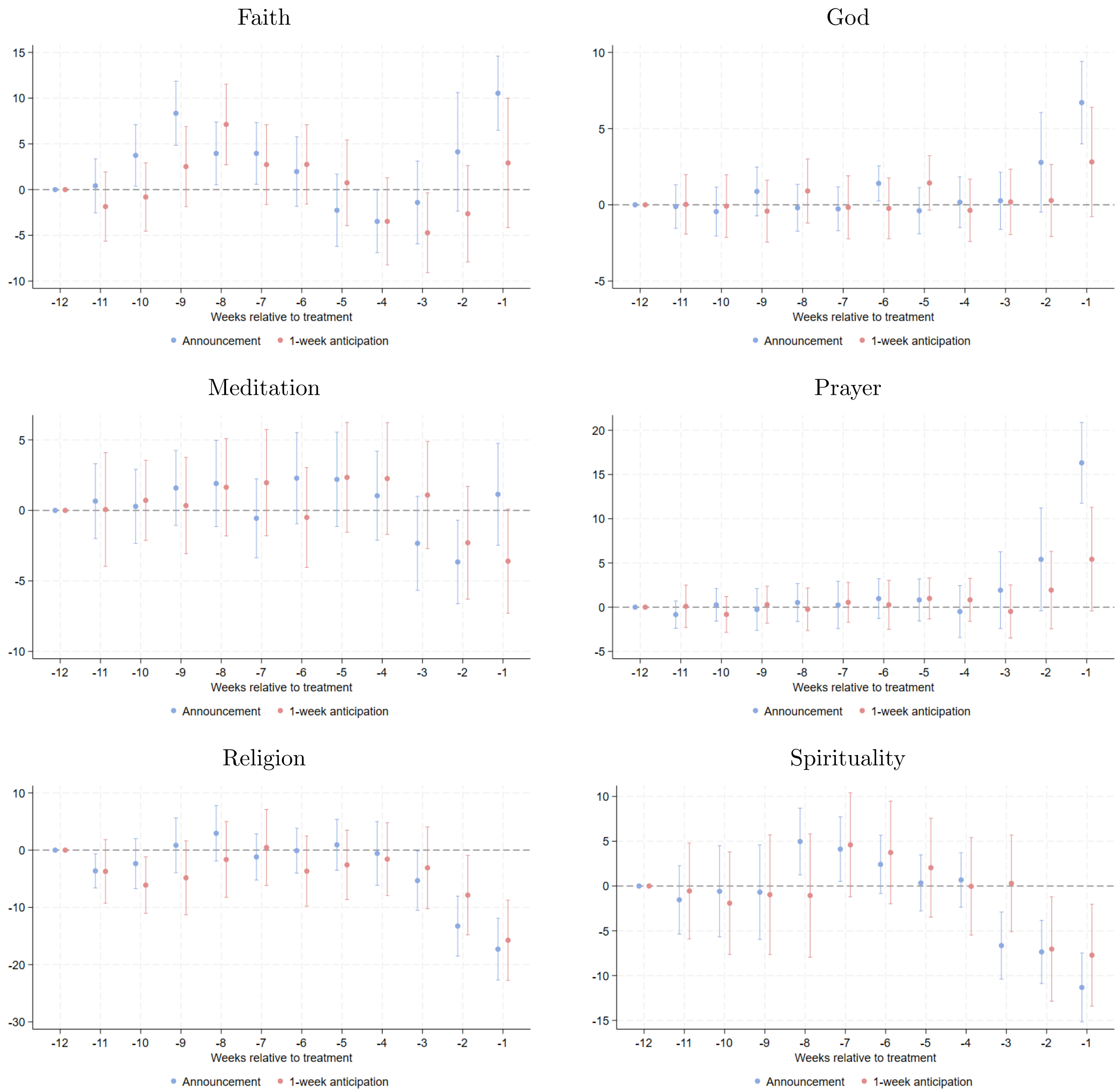
Notes: This figure shows the evolution for each topic using the rescaled series for 2019 (blue) and 2020 (red). Relative days to lockdown announcement are in the x-axis. For 2019 these correspond to the paired days.

Figure 3: TWFE estimates and Covid-19 controls



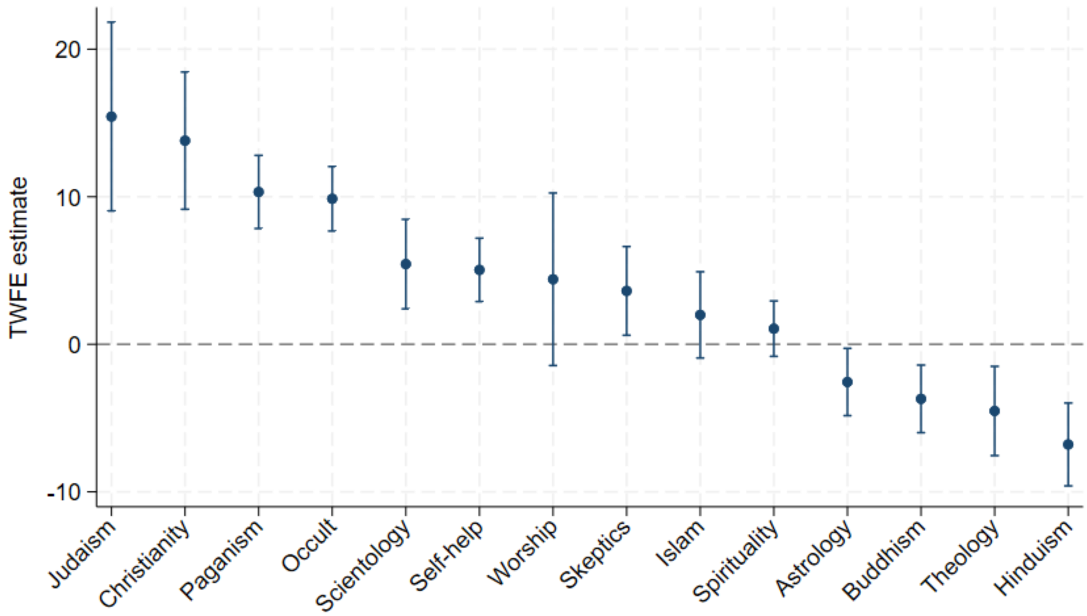
Notes: This figure compares TWFE estimates as in Equation 1 both including and not including one day lagged number of confirmed COVID-19 deaths. Confidence intervals at the 95% level.

Figure 4: Pretrends test: Topics



Notes: This figure reports the pre-trends test coefficients as in 2. Blue lines use the lockdown announcement day as beginning of treatment, whereas as red lines redifine treatment to begin 7 days before announcement. Confidence intervals at the 95% level.

Figure 5: TWFE estimates on Subcategories



Notes: This plots TWFE estimates of each subcategory withing the 'Religion & Belief' category. Confidence intervals at the 95% level.